





14

==

1







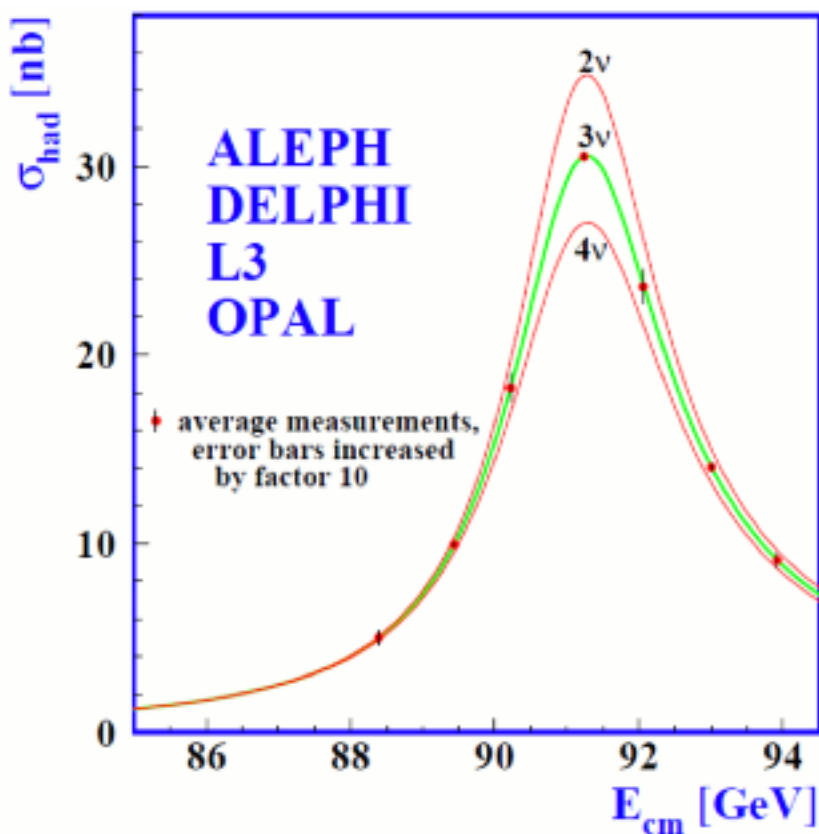
















# Flavors

👉 Three flavors of neutrino are discovered:

- $\nu_e$  (electron, 1956)
- $\nu_\mu$  (muon, 1962)
- $\nu_\tau$  (tau, 2000)

👉 Three lepton numbers are conserved for massless neutrino:

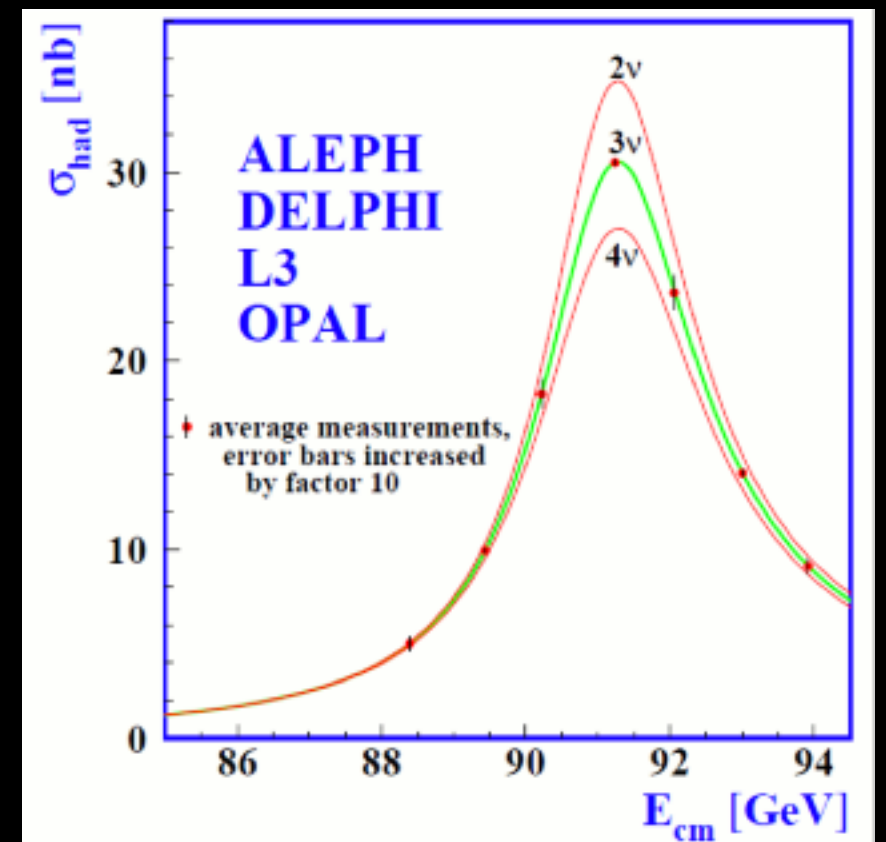
✅ Allowed:

$$\nu_e + n \rightarrow p + e^-$$

❌ Forbidden:

$$\nu_\mu + n \not\rightarrow p + e^-$$

$$\Gamma(Z \rightarrow \text{all})$$



Only three (light) neutrino types possible

# Mass